

Article

Comprehensive Properties of Grafted Polypropylene Insulation Materials for AC/DC Distribution Power Cables

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Abstract: Polypropylene (PP) exhibits excellent insulation properties, high thermo-stability, and recyclable nature, thus expected to be the next-generation insulation material for HV cable application. Chemical grafting modification is an effective technology to improve the electrical properties of polypropylene. In this paper, we develop and report a new-type grafted PP insulation material by water-phase grafting technology. The comprehensive properties including electrical, thermal, and mechanical of it are tested and compared with traditional cable insulation material—crosslinked polyethylene (XLPE). The results show that the grafted PP holds superior thermal properties and enough mechanical flexibility. The electrical properties are of significant advantages, including resistivity enhanced by nearly two degrees of magnitudes, AC/DC breakdown strength raised by over 20%, and obviously suppressed space charge accumulation. These results indicate that grafted PP is very suitable for application in HV cable systems, either AC or DC. This research lays a foundation for the research and development of the next-generation recyclable polypropylene cables at higher voltage levels.

Keywords: grafted polypropylene; recyclable insulation materials; properties



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1. Introduction

Power cables will become mainstream in urban power transmission and distribution, long-distance trans-sea power transmission, far offshore wind power, and onshore new energy power transmission. With the low-carbon and environmental protection development of power equipment, it is urgent to develop cable insulation materials with recyclable nature, high economical efficiency, and high thermostability [1,2]. At present, cross-linked polyethylene (XLPE) is widely used as power cable insulation material, but XLPE cannot be recycled after the end of the cable lifetime, and the manufacturing processes of XLPE cables include cross-linking and degassing, which increases energy consumption and carbon emissions. Therefore, in order to reduce the carbon emission of the whole life cycle and realize recycling after the end of cable lifetime, polypropylene (PP) has drawn great attention from both academia and industry because of its superior electrical properties, high thermostability, and recyclable nature, as shown in below Table 1 [3–5].

The modification method of PP mainly consists of blending, copolymerization, nanodoping, and grafting modification. The blending is the PP matrix mixing with other polymer matrices. The copolymerization modification is propylene monomer copolymerizing with other olefin monomers. The most commonly used blend/copolymerization routes of PP insulation include polypropylene/elastomer blends [6], ethylene-propylene copolymers [7], ethylene-octene copolymers [8], which can reduce the elastic modulus

of PP, whereas also the electrical insulation properties. The nanodoping refers to doping a certain amount of nanoparticles into a PP matrix, such as MgO [5], ZnO [9], SiO₂ [10], which can significantly improve the electrical insulation properties, but the agglomeration of nanoparticles in PP matrix cannot be completely avoided, it restricts the feasibility of its large-scale production [11]. The grafting modification is to graft the chemical groups onto the molecular chains of PP, which is a modification method at a microscopic level to realize the improvement of macroscopic electrical properties. The grafted groups can introduce a deep trap into the material, thus improving the electrical insulation properties [12]. Furthermore, compared with nanodoping, grafting modification has no agglomeration, which is more suitable for large-scale industrial preparation and production.

Table 1. Comprehensive properties of PP versus XLPE [5].

Properties	PP	XLPE
DC volume resistivity ($\Omega \cdot m$)	1.7×10^{15}	0.9×10^{15}
DC breakdown strength (MV/m)	399	300
Operation temperature ($^{\circ}C$)	Over 90	70–90
Thermal nature	Thermoplastic	Thermoset
Mechanical property	Hard and brittle	Soft and flexible

To evaluate the comprehensive performances of grafting modification, this paper carries out an experimental study on the comprehensive properties of grafted polypropylene insulation material for AC/DC distribution power cables. The AC and DC electrical, thermal, and mechanical properties of grafted PP material were tested and analyzed. We also selected two mature XLPE cable insulation materials as references to make a direct comparison. This work reveals the pros and cons of PP-based and XLPE insulation materials, and it would provide a valuable reference for the research and development of next-generation recyclable polypropylene cables at higher voltage levels.

2. Materials Preparation and Characterization

2.1. Materials Preparation

The styrene grafted polypropylene was marked as PPg, which was jointly developed by Tsinghua University and Sinopec (Beijing) Chemical Industry Research Institute. The PP matrix material used for PPg preparation was from Sinopec, and the brand name was NS20. The molecular weight is about 370,000 g/mol. Water-phase grafting technology was adopted for the preparation. Benzoyl peroxide (BPO) was used as the initiator in the reaction process. The free radical formed by heat decomposition of the initiator, had taken the hydrogen atoms of PP to form PP macromolecular radical. The PP macromolecular radical resulted in a grafting reaction with the styrene monomer, as shown in Figure 1 [11]. The reaction of this technology is mild and easy to control, simple operation, of few residues, high purity, and has the feasibility of large-scale preparation and industrial production. For comparison, two commercial XLPE cable insulation materials in distribution lines were selected and produced by Zhejiang Wanma Macromolecule Material Group Co., Ltd. (Hangzhou, China) and Shanghai New Shanghua Polymer Material Co., Ltd. (Shanghai, China), the card number YJ-35 and 4205EC-35 marked as XLPE1 and XLPE2 respectively.

By plate vulcanizer pressing, the three materials PPg, XLPE1, and XLPE2 were prepared into film samples with the required thickness for each test. The XLPEs were first pressed for 10 min at 120 $^{\circ}C$ and 15 MPa, then crosslinked for 15 min at 180 $^{\circ}C$, during which the polyethylene can chemically crosslink into a polymeric network. Finally, it was cooled to room temperature through circulating water. The PPg was pressed for 10 min at 200 $^{\circ}C$ and 15 MPa, finally cooled to room temperature in the same manner. The prepared samples were left in a vacuum oven for 24 h at 70 $^{\circ}C$.

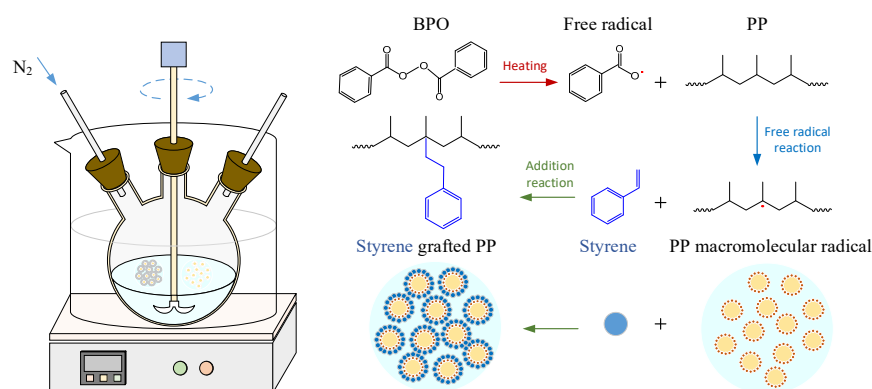


Figure 1. The reaction of styrene grafted PP.

2.2. Materials Characterization

The Fourier transform infrared spectrometer (FTIR) was used for chemical group characterization. Transmission mode was used with the scanning number of 32 and resolution of 4 cm^{-1} , and scanning range from 400 cm^{-1} to 4000 cm^{-1} . To eliminate the interference of unreacted and self-polymerized grafting monomer, the Soxhlet extraction was adopted to remove the ungrafted styrene. The prepared grafted PP was extracted by ethyl acetate for 24 h, then dried and prepared into a film sample as described in Section 2.1. Then the FTIR absorption spectra of the purified samples were tested. Besides, to identify the characteristic peaks of grafted PP, pure PP without any modification is also prepared into a film sample and the FTIR is tested [13].

The Differential scanning calorimeter (DSC) was used for heat flow characterization. The samples were heated to $200\text{ }^{\circ}\text{C}$ and maintained for 5 min, then cooled to $30\text{ }^{\circ}\text{C}$ at the rate of $10\text{ }^{\circ}\text{C}/\text{min}$ for testing the hot flow in the crystallization process. And the samples were re-heated to $200\text{ }^{\circ}\text{C}$ at the rate of $10\text{ }^{\circ}\text{C}/\text{min}$ for testing in the melting process.

The universal tension tester was used to test the tensile properties. The samples were dumbbell shaped with an average thickness of $200\text{ }\mu\text{m}$, parallel part length of 33 mm , narrow width of 6 mm , and a tensile rate of $20\text{ mm}/\text{min}$. Each kind of material was tested 5 times, and the measurement results were the mean and standard deviation values.

The three-electrode resistivity test platform was used for DC volume resistivity testing, and the DC leakage current was measured by an electrostatic current meter (Keithley 2635b, USA). The average thickness of the sample was $100\text{ }\mu\text{m}$. The test temperature was $30\text{ }^{\circ}\text{C}$ and $70\text{ }^{\circ}\text{C}$, respectively, and the electric field strength ranged from $5\text{ kV}/\text{mm}$ to $60\text{ kV}/\text{mm}$ with the interval of $5\text{ kV}/\text{mm}$. The current values in the stable segment were averaged as the measured value.

The thermally stimulated depolarization currents (TSDC) were used for testing the trap energy level distribution of materials [14]. The average thickness of the sample was $100\text{ }\mu\text{m}$. The samples were applied electric field strength of $5\text{ kV}/\text{mm}$ at a polarization temperature of $70\text{ }^{\circ}\text{C}$ for 30 min, then short-circuited for 5 min after rapidly cooled to $-60\text{ }^{\circ}\text{C}$, and finally heated to $120\text{ }^{\circ}\text{C}$ at the rate of $3\text{ }^{\circ}\text{C}/\text{min}$ meanwhile measuring the TSDC.

The HVDC tester was used to test the DC breakdown strength, and the HVAC tester was used to test the AC breakdown strength. The samples with an average thickness of $100\text{ }\mu\text{m}$ were placed between the sphere-sphere electrodes. The electrodes and the sample were immersed in silicone oil for preventing the flashover. The sample was applied voltage at the rate of $1\text{ kV}/\text{s}$ until breakdown. The measured values of 15 different breakdown locations were selected in each sample, and statistical analysis was adopted by the two-parameter Weibull distribution [15].

The broadband dielectric spectrometer (BDS), modeled as Novocontrol Concept 80, was used for testing dielectric properties. The average thickness of the sample was $100\text{ }\mu\text{m}$. The sample was applied 1 V_{rms} , the frequency from 1 Hz to 10^6 Hz , the temperature of $30\text{ }^{\circ}\text{C}$ and $70\text{ }^{\circ}\text{C}$ respectively.

The pulsed electro-acoustic method (PEA) was used for testing the change of space charge. The average thickness of the sample was 200 μm . The sample was applied a polarized electric field of 30 kV/mm at room temperature, and a polarization time of 40 min.

3. Results and Discussion

3.1. FTIR

The results of infrared spectroscopy were shown in Figure 2. It can be seen that compared with pure PP, new vibration peaks of the benzene ring skeleton of PPg appeared at wavenumbers of 1500 cm^{-1} and 1600 cm^{-1} , as shown in the yellow box. The three C-H bond expansion peaks in benzene rings appeared from 3105 cm^{-1} to 3020 cm^{-1} , as shown in the green box. The vibration peak of the benzene ring skeleton appeared at 700 cm^{-1} , as shown in the blue box. It indicated that the purified PPg material was still able to characterize the characteristic peaks of the styrene groups, which ensured the successful grafting of the styrene groups on PP molecular chains.

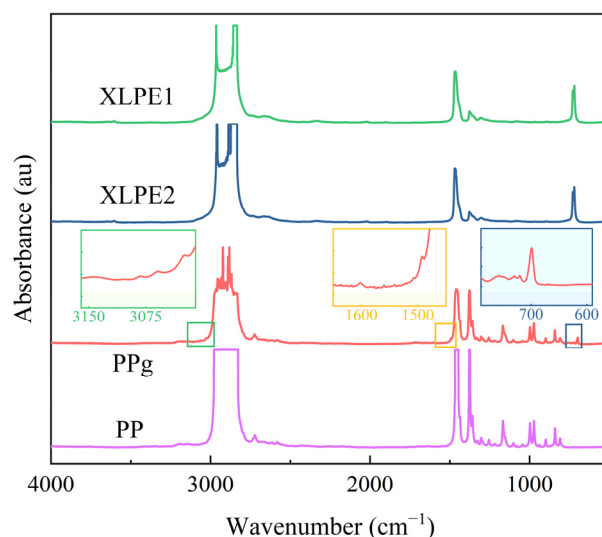


Figure 2. FTIR of the three materials.

3.2. DSC

The melting and crystallization heat flow of the three materials were obtained by the DSC test as shown in Figure 3. The thermal parameters of the materials were shown in Table 2. The crystallinity was calculated by dividing the measured melting enthalpy by 100% melting enthalpy, where 100% melting enthalpy of XLPE1 and XLPE2 was 290 J/g, and that of PPg was 209 J/g [15]. As shown in Figure 3 and Table 2, the melting peak temperature of PPg was significantly higher than that of XLPE1 and XLPE2 by approximately 50 $^{\circ}\text{C}$, and XLPEs began to absorb heat above 85 $^{\circ}\text{C}$, which limited the increase of the current-carrying capacity of the cables. The PPg started heat absorption above 135 $^{\circ}\text{C}$, and the melting peak temperature exceeded 150 $^{\circ}\text{C}$ and was approximately 50 $^{\circ}\text{C}$ higher than XLPEs. Because polyethylene (PE) was a semi-crystalline plastic, the crystallinity of XLPE decreases with the increase of crosslinking degree. The crystallinity of PP was generally lower than that of PE, in addition, the steric hindrance effect produced by the grafted benzene ring groups hindered the movement and order of PP molecular chains to a certain extent, and finally decreases the crystallinity.

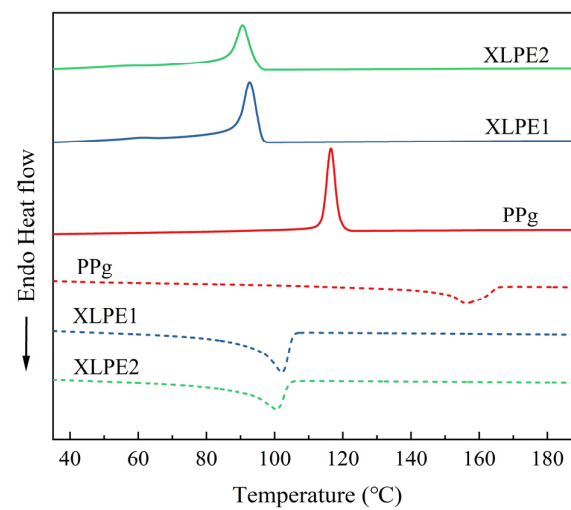


Figure 3. DSC results of three materials.

Table 2. Thermal parameters of three materials.

Thermal Parameters	XLPE1	XLPE2	PPg
The temperature of melting peak (°C)	102.2	100.6	156.3
Temperature of crystallization peak (°C)	92.7	90.6	116.5
Melting enthalpy (J/g)	104.1	102.4	49.9
crystallinity (%)	36.0	35.3	23.9

3.3. Mechanical Tensile Test

The stress-elongation curves obtained by the mechanical tensile test are shown in Figure 4, and the corresponding parameters of mechanical properties are listed in Table 3. As shown in the subplot of Figure 4, the overall tension modulus of PPg was higher than that of XLPEs, due to the weaker intrinsic toughness of the PP matrix above the glass transition temperature. The PPg had no thermosetting cross-linking structure and failed to form a network structure, which made the PP molecular chains more extended. In addition, the grafted styrene groups with a large volume, reduced the intermolecular binding forces, and the molecular chains were more likely to slip, making the break elongation slightly higher than XLPEs. The PPg with low crystallinity, made the reduction of macromolecules subjected to stress during stretching, it also increased the break elongation. In a word, the mechanical tensile properties of PPg could meet the requirements of minimum break elongation of 200% and minimum tensile strength of 12.5 N/mm² in cable insulation [16].

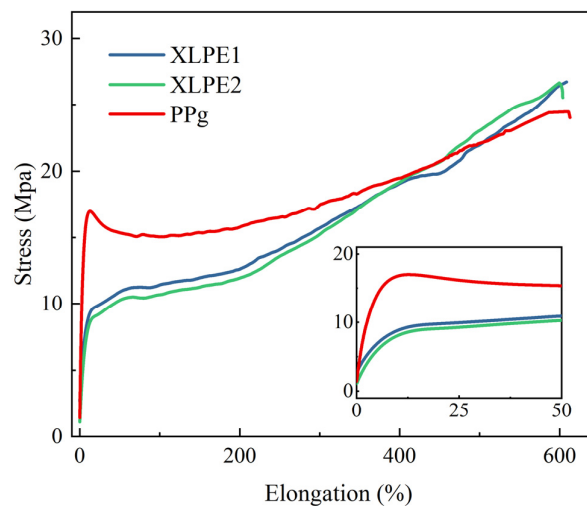


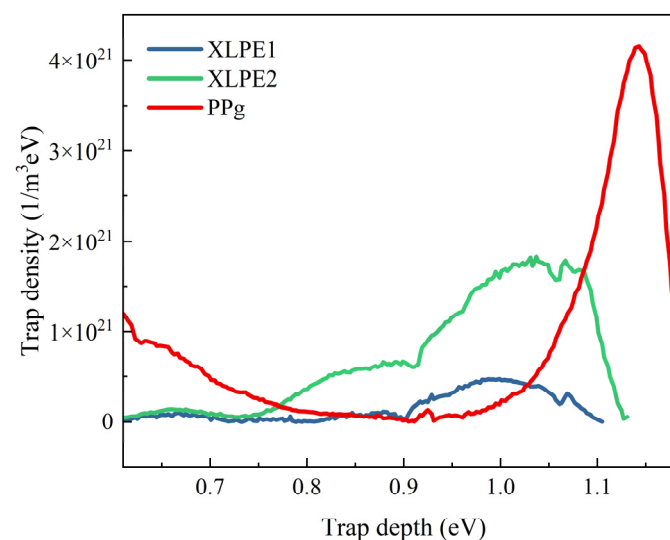
Figure 4. Stress-elongation curves of three materials.

Table 3. Mechanical properties of three materials.

Mechanical Properties	XLPE1	XLPE2	PPG
Tension modulus (MPPa)	120.9 ± 7.4	127.5 ± 7.3	538.3 ± 45.5
Elongation (%)	555.6 ± 19.7	622.1 ± 28.3	652.7 ± 32.3
Tensile strength (N/mm ²)	28.8 ± 7.1	32.5 ± 1.9	31.8 ± 2.1

3.4. TSDC

The thermal stimulation current test obtained the change of current with temperatures, and the relationship of trap density and depth was calculated by the improved TSDC analysis method, the result of which is shown in Figure 5 [14]. The trap depth of PPg was 1.145 eV, XLPE1 of 0.995 eV, and XLPE2 of 1.031 eV. The results showed that PPg had deeper traps and higher density. This is due to the grafted styrene groups introducing aromatic ring structure into the PP molecular chains. The aromatic ring possesses delocalized π bond, which is an electron-affinitive structure, therefore it can act as a deep charge trap, meanwhile changing the electronic state distribution. Moreover, the grafted styrene groups affected the crystallization behaviors of PP, the size of the spherulites became smaller and the boundary became blurred. Thus, it increased the interface between the crystalline region and the amorphous region and made the trap quantity increase. A large increase in deep trap density could capture more carriers and reduce the migration of charges, which was conducive to improving the resistivity and breakdown strength [12].

**Figure 5.** Trap distribution of three materials.

3.5. DC Volume Resistivity

The results of DC volume resistivity were shown in Figure 6, the applied electric field strengths of 5 kV/mm, 15 kV/mm and 25 kV/mm, and the temperatures at 30 °C and 70 °C. The DC volume resistivity of XLPEs, at the high electric field and high temperature, reduced by 2 orders of magnitude from that at low field and high temperature. The volume resistivity of PPg was higher than XLPEs at different field strengths and temperatures, and the reduction rate of that was much lower than XLPEs with the field strength and temperature increasing. It indicated that the volume resistivity of PPg was less dependent on the field strength and temperatures. The higher volume resistivity of PPg could be attributed to the grafted styrene groups introducing deep traps, which captured the carriers and reduce the mobility of carriers [5,11], thereby reducing the leakage current and improving the resistivity, corresponding to the trap level distribution in Figure 5. This indicated that the DC leakage loss of PPg was lower than that of XLPEs at high temperatures, which were

beneficial for the safe and stable operations of cables at conditions of high temperatures and high field strength.

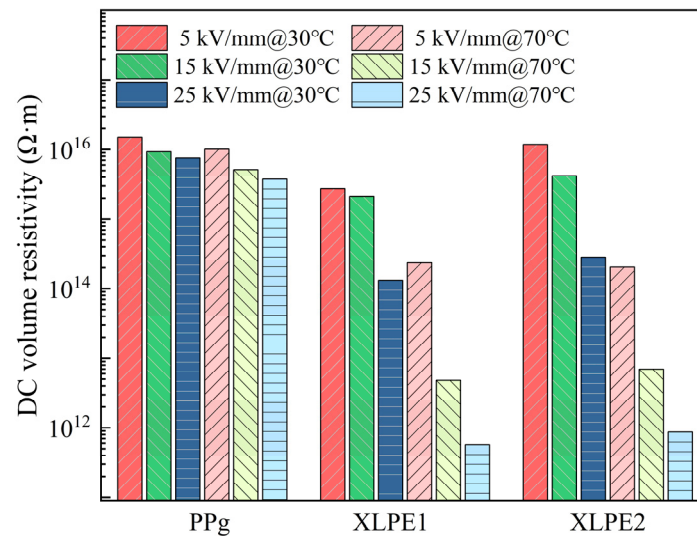


Figure 6. DC volume resistivity of three materials.

3.6. AC and DC Breakdown Strength

The AC and DC breakdown strength can be analyzed by Weibull distribution, as shown in Equation (1):

$$P(E) = 1 - \exp\left(-\left(\frac{E}{E_C}\right)^\beta\right) \quad (1)$$

where $P(E)$ is the cumulative breakdown probability, and E is the experimental breakdown strength value. E_C is the fitted characteristic breakdown strength, which corresponds to the electric field strength at the cumulative breakdown probability of 63.2%. The fitted breakdown strength curves of the three materials are shown in Figures 7 and 8, respectively. While the E_C values are shown in Figure 9. The AC breakdown strength of PPg was higher than XLPEs, and it was approximately 34.2% higher than XLPE2 at ambient temperature, and approximately 38.9% higher than XLPE2 at high temperature. The DC breakdown field strength of PPg was also higher than XLPEs, and it was approximately 20.9% and 23.0% higher than that of XLPE1 and XLPE2 at ambient temperature, and approximately 105.2% and 132.3% higher at high temperature respectively. It indicated that PPg still maintained strong DC breakdown resistance at high temperatures, and the breakdown value was close to that of XLPEs at ambient temperature. Meanwhile, the DC breakdown strength of the three materials was higher than the AC breakdown strength of that. Concurrently, it had been discovered that the DC breakdown field strength of the three aforementioned materials exceeded their AC breakdown field strength. This phenomenon was attributed to the homopolar injection of space charges, which provoked gradual movement towards the interior of materials and eventual breakdown. The homopolar injection effectively space charges inhibited further charges injecting. Conversely, during AC breakdown, the alternating electric field resulted in charge injection accumulation at the interface between the medium and electrode, leading to a distortion in local field strength. Furthermore, the repeated trapping and detachment within the materials generated radiation energy, inducing molecular chain breakage and high-energy electron collision ionization, ultimately increasing the current carriers and resulting in AC breakdown [17].

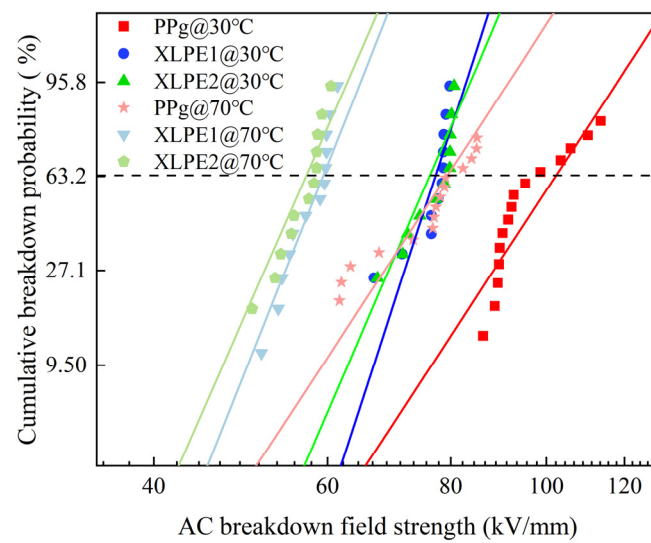


Figure 7. The Weibull distribution of AC breakdown field strength, where the lines are the corresponding fitting curves of Weibull distribution.

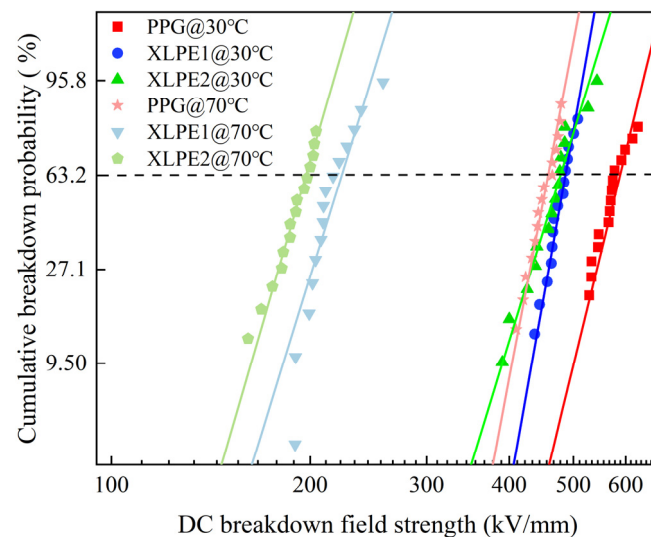


Figure 8. The Weibull distribution of DC breakdown field strength, where the lines are the corresponding fitting curves of Weibull distribution.

Due to the grafted styrene groups in PPg having introduced deep traps (as shown in Figure 5), it captured the charge carriers during the breakdown process, greatly reduced the kinetic energy and mobility thus suppressing the intermolecular energy exchange, thus increasing the DC breakdown field strength [12]. The improvement of AC breakdown field strength, on the one hand, can also be attributed to the introduced deep traps, on the other hand, grafting the rigid styrene groups formed an entangled structure, which enhanced the interaction of molecular chains and thus improved the breakdown properties in high electrical and high thermal fields. It showed that PPg material had excellent breakdown resistance for AC/DC general cable insulation.

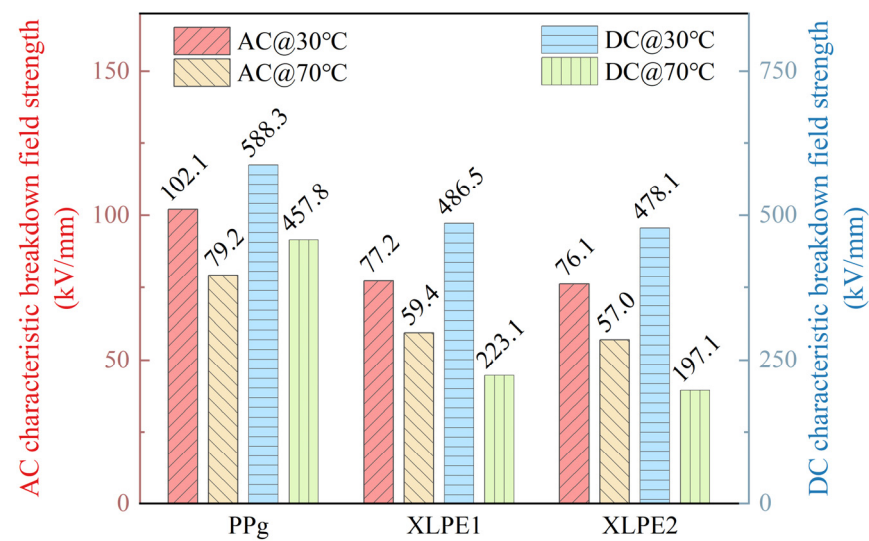


Figure 9. AC and DC characteristic breakdown field strength.

3.7. Dielectric Properties

The present study investigated the frequency-dependent changes in the relative permittivity and dielectric loss tangent ($\tan \delta$) of three materials, as shown in Figures 10 and 11. The results indicated that the relative permittivity of the materials exhibited a frequency-dependent behavior and decreased with an increase in temperatures, albeit with a marginal shift in the curve. The rise of temperatures did not alter the general shape of the $\tan \delta$ curves, but only caused a rightward shift along the frequency axis. Notably, the $\tan \delta$ value of PPg displayed an initial increase followed by a decrease with an increase in frequency, with a crossing point observed around the power frequency of 50 Hz. In the high-temperature and low-frequency range (below 50 Hz), the $\tan \delta$ value of PPg was lower than its corresponding ambient temperature value, while it was higher than the ambient temperature $\tan \delta$ value in the medium-high frequency range (above 50 Hz). Similarly, the $\tan \delta$ of XLPEs increased initially and then decreased with frequency, with the peak of the curve observed at a frequency level of 10^4 Hz.

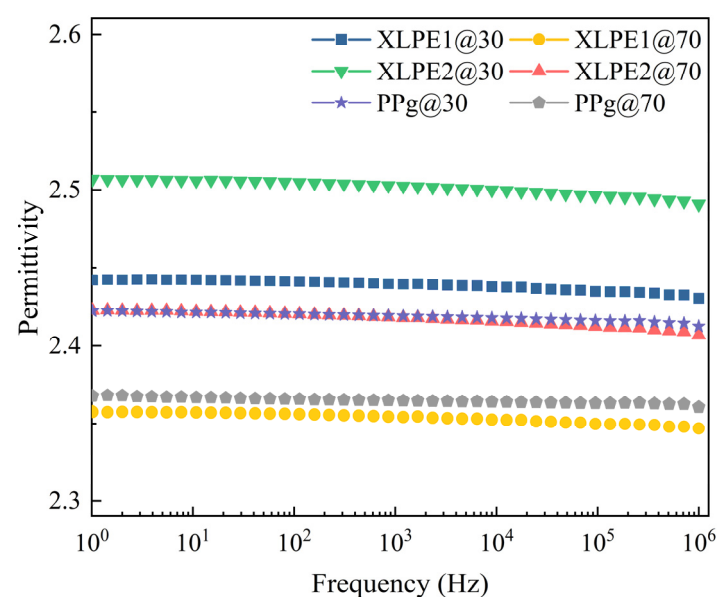


Figure 10. Permittivity varies with frequency.

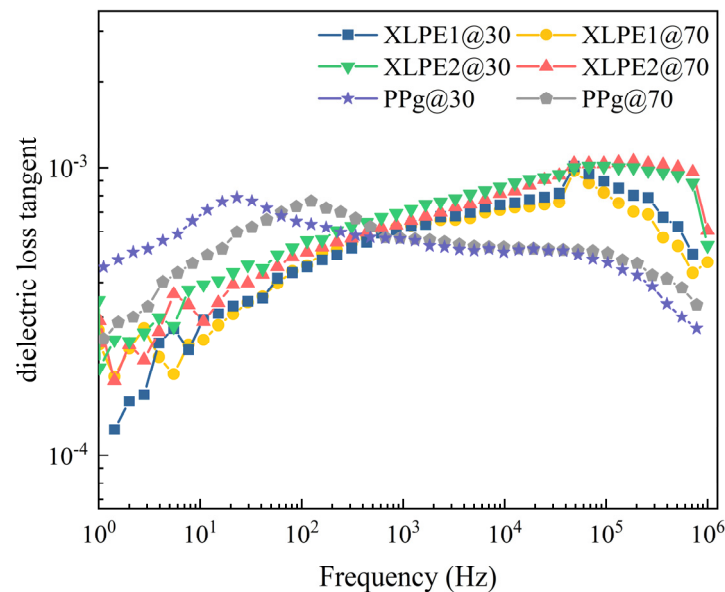


Figure 11. Dielectric loss tangent varying with frequency.

The observed dielectric loss in the materials was attributed to the conduction and polarization losses. The presence of styrene groups grafted in PPg enhanced the orientation and interfacial polarization losses in the low-frequency range. This results in a slightly higher $\tan \delta$ value for PPg compared to XLPEs in this frequency range, with the difference increasing with frequency. In the medium-high frequency range, the interfacial polarization strength weakened, and the dipole orientation polarization gradually dominates the dielectric response behavior. The entanglement effect of the grafting chains formed by the styrene groups suppressed the movement of the molecular chains, thereby weakening the dipole orientation polarization and causing a significant decrease in the $\tan \delta$ value of PPg in the high-frequency range [18]. At high temperatures, the increased disorder and intense movement of molecular chains made it challenging to establish polarization, resulting in a decrease in relative permittivity and dielectric loss. As shown in Figure 11, the relative permittivity of PPg at 30 °C and 70 °C was approximately 2.42 and 2.36, respectively, with $\tan \delta$ values at 50 Hz of 7.02×10^{-4} and 6.79×10^{-4} . The $\tan \delta$ value of XLPEs exhibited less temperature dependence, and the collision ionization generated with increasing frequency increased the carriers, resulting in an increase in the $\tan \delta$ value of XLPEs.

3.8. Space Charge

It illustrated the distribution of space charges density and electrical field distribution in the thickness direction of the three materials at ambient temperature in Figure 12. Among them the electric field distribution is calculated by Poisson Equation (2) [5]:

$$E(x) = \frac{1}{\epsilon_r \epsilon_0} \int_0^d \rho(x) dx \quad (2)$$

where ϵ_0 is the vacuum permittivity, and the ϵ_r is the relative permittivity of the sample. d is the thickness of the sample. $\rho(x)$ is the distribution of space charge obtained by PEA, while $E(x)$ is the electric field distribution. It was not observed the significant accumulation of space charges or the distortion of the electrical field in PPg material. However, XLPE1 showed a small accumulation of the same polarity charges, which intensified the field distortion near the electrode. On the other hand, XLPE2 showed a small accumulation of opposite polarity charges, and the degree of field distortion was slightly lower than that in XLPE1 due to the charge's recombination. The accumulation of opposite polarity charges in XLPEs was attributed to by-products generated during the cross-linking reaction, while the accumulation of the same polarity charges was mainly caused by electrode injec-

tion [19]. The styrene groups grafted in PPg exhibited an excellent ability to suppress the accumulation of space charges and field distortion. It was due to the introduction of a large number of deep traps after the grafting styrene groups, which captured carriers to reduce carriers mobility. Additionally, the captured charges can form a local potential barrier which can suppress the following injection of electrode charges. The ability to suppress the accumulation of space charges could prevent local degradation problems caused by charges accumulation in DC cable insulation, thereby improving the safe operation of DC cables.

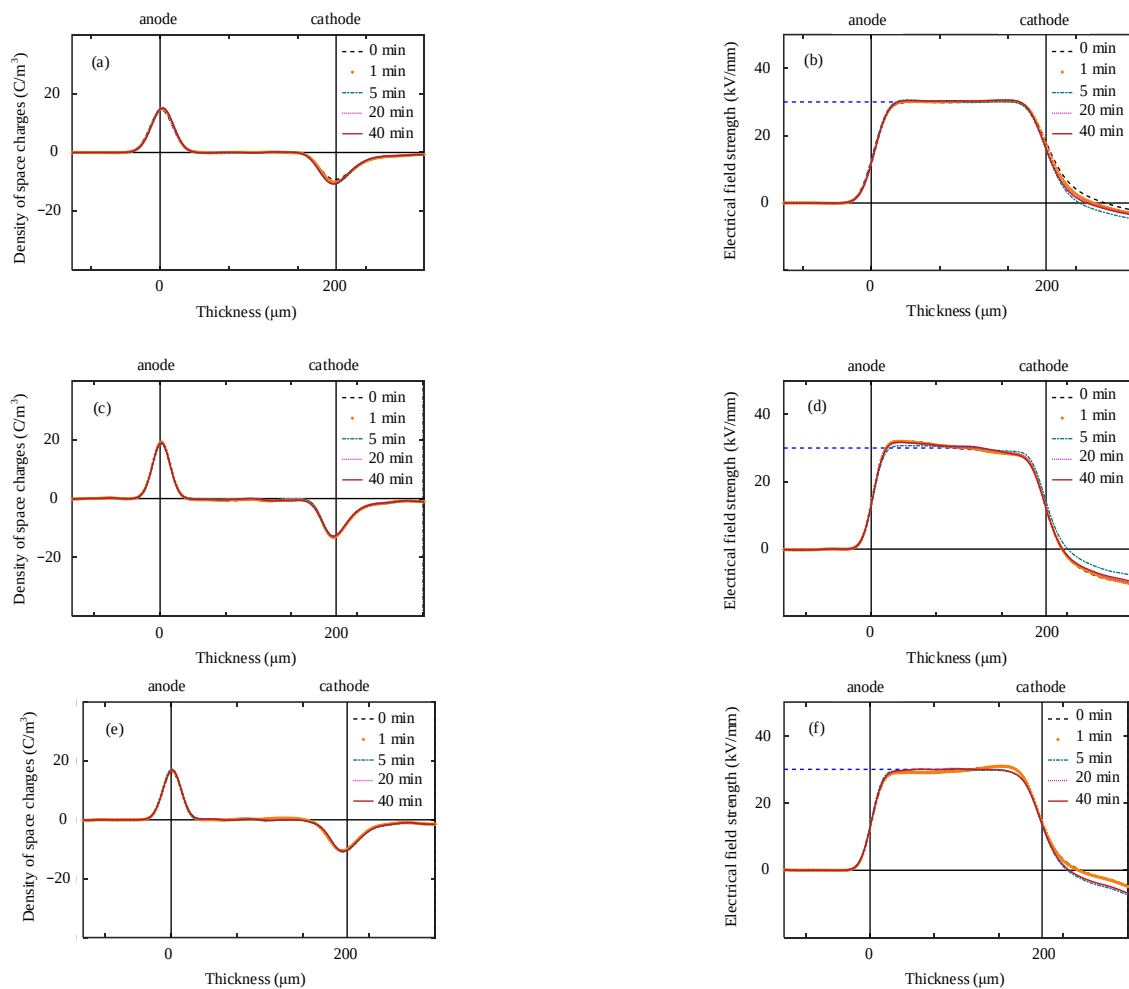


Figure 12. Characteristics of space charges. (a–f) are the density of space charges and electrical field strength in PPg, XLPE1, and XLPE2, respectively.

4. Conclusions

It was studied that grafted polypropylene insulation materials for AC/DC distribution power cables, and their comprehensive properties were evaluated. The results demonstrated that:

(1) The PPg has possessed superior AC/DC electrical properties compared to XLPE materials, including higher DC volume resistivity at high temperatures, higher breakdown field strength, and a better suppression of space charges. The volume resistivity of PPg at high temperatures was found to be 1 to 2 orders of magnitude higher than that of XLPEs. Additionally, the characteristic breakdown field strength of PPg was higher than that of XLPEs, with the DC characteristic breakdown field strength of PPg at high temperatures being over 100% higher than that of XLPEs. The permittivity and dielectric loss of PPg at high temperatures were slightly lower than those at ambient temperature, and showed different dielectric properties from XLPEs. Furthermore, the degree of space charge accumulation and electric field distortion in PPg were weaker than that in XLPEs. These findings

indicated that PPg exhibited excellent material performances in AC/DC distribution power cables at high temperatures and high field conditions.

(2) The PPg has possessed weaker mechanical properties compared to XLPE materials, and with a higher tension modulus. Further research is needed to focus on optimizing the matching of mechanical and electrical properties. However, the mechanical properties of PPg were found to meet the relevant standard requirements for the operation of power cables.

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